Adaptability

One Operation Behind Shape Matching and Analogy, and Why It Is Not Called Intelligence

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Draft — draft-2026-08-30.4
1 September 2026 • document revision 3

DRAFT — draft-2026-08-30.4 r3 (1 September 2026). Circulated for comment, not as a finished result. This is the theory half of a pair; the companion reports what is built and what is not. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

Two capabilities this programme treats separately — matching a shape against a held codebook, and reasoning by analogy — are one operation at two grains. Both *extract the transform between things already held and apply it where nothing is held.* Naming that operation gives a quantity worth having: *adaptability*, the capacity to acquire a competence that was not supplied. The word is chosen against *intelligence* deliberately. Intelligence is a scalar people rank; adaptability is an operation with a cost, an ordering, and a failure mode, and a system either performs it on a given case or does not.

1 Why not intelligence

Intelligence names a ranking before it names a mechanism. Asking whether a system has more of it than another invites a scalar, and a scalar invites a benchmark, and a benchmark measures what a system already does rather than what it can come to do. That is the wrong quantity for the question here.

Adaptability is narrower and is an operation rather than a rank. A system is adaptable on a case to the degree that, meeting something it does not hold, it ends up holding it — and the interesting part is the mechanism by which the new holding is *derived from* the old rather than supplied. A system given a competence has not adapted. A system that recognises everything it meets has not adapted either; it was already sufficient. *Adaptation requires a residual*, and the size of what must be added to absorb one is the cost.

2 The operation

Definition 2.1 (Adaptation)**.** *Given a held set and an observation not covered by it, an* adaptation *is the addition of the smallest structure under which the observation is covered, derived from what is held.*

Three words carry it. *Smallest*, because an adaptation that adds the observation itself is storage, not adaptation. *Derived*, because an adaptation supplied from outside is instruction. And *covered*, because the test is whether the case is now recognised, not whether a description of it exists.

Remark 2.2 (Shape matching and analogy are one operation). Matching a shape against a codebook and completing an analogy look unrelated because their vocabularies are. They are not.

To decode an observation against held signatures is to ask what transform carries a held one onto it. Where that transform is the identity, the observation is recognised. Where it is a uniform step along a dimension the codebook already has, the observation is a known thing moved — and no new structure is required. Where no such step exists but the residual is *coupled* across the observed dimensions, the transform requires a dimension not currently held, and one must be inferred before the observation can be covered.

To complete an analogy is the same question with the arguments rearranged. Given two things held, the transform between them is extracted; given a third, that transform is applied to reach a fourth which is not held. *Anti-unification — generalising two structures while retaining what distinguishes them — is the general form, and the shape ladder is that form restricted*: at one rung the difference is a single step along a known dimension, at the next it is a dimension.

So the two are not analogous to each other. They are the same operation, and the second is what the first looks like when the objects are rules rather than observations.

3 The ladder, and why its order is not arbitrary

An adaptation must be the smallest available, and that gives an ordering rather than a preference:

1. **Recognise.** The observation is covered as held. Nothing is added, and no adaptation has occurred — which is the correct outcome for most observations.

2. **Transform in the held basis.** One uniform step along a dimension already carried. The observation is a known thing displaced; the codebook does not grow.

3. **Infer a dimension.** The residual is coupled across observed dimensions in a way no step in the current basis produces, so a dimension is minted and the transform is expressed in it.

4. **Mint.** No transform reaches the observation, so it is added as new.

Only rungs three and four are adaptations, and only the third is interesting: it is where the system's own vocabulary changes. Rung four is the honest floor — something genuinely new — and a system that reaches it too readily is not adapting but accumulating.

Remark 3.1 (Why the order must be Occam and not preference). Attempting mint first would always succeed, so a system that did would never transform, never infer, and would grow linearly in what it meets. Attempting transform first is what makes minting *informative*: a mint that survives the earlier rungs is evidence that nothing held reaches the case, which is a claim about the held set and not only about the observation.

That is also why the ordering is a commitment and not an implementation detail. *A system that mints where a transform would have sufficed has recorded a coincidence as a category*, and one that transforms where a mint was required has folded a distinction it needed. Both are failures of adaptation, in opposite directions, and the ladder is what makes them separable.

4 What adaptability predicts

The account is worth having only if it says what an adaptable system does that an unadaptable one does not, on a case neither has seen.

One mint, then recognition. Meeting a genuinely novel case repeatedly, an adaptable system should mint *exactly once* and recognise every later instance — not mint per encounter, which is storage, and not fail to mint, which is refusal. The signature is a step followed by silence.

Zero mints under transformation. Meeting a known thing transformed along a held dimension, it should add nothing at all. A system that mints there is treating the dimension it already has as absent.

Coupling forces dimensions. A residual that is coupled across observed dimensions should produce a dimension, not a member. *The distinguishing case is the one whose difference cannot be written in the current basis at all*, and a system without rung three cannot tell it apart from novelty — it mints where it should have generalised, and the codebook grows with the world instead of with its structure.

And competence should transfer without being carried. If the operation is real, structure built for one domain should be usable unmodified in another whose existence was not anticipated when the structure was made. That is the strongest available test, because it removes the author from the loop, and it is the one the assessment companion reports against.

5 What this does not claim

Adaptability is not a quantity a system has more or less of in general. It is defined *per case* and against *a held set*, so the same system is adaptable on one observation and not on another, and moving the held set moves the verdict. There is no frame-free adaptability, for the same reason this programme's base claim gives for there being no frame-free anything.

Nor does adaptation imply improvement. An adaptation is an addition that covers a residual; whether covering it was worth the cost is a separate question with a separate answer, and a system that adapts readily to noise is worse than one that refuses.

6 A domain where it did not carry the load

The account would be worth less if it were only presented where it holds, and the companion programme contains a case that cuts against it. It is reported here rather than there because a theory should carry its own counter-evidence.

In program synthesis the machinery above is present: a held codebook proposes, an enumerator searches, and an execution verdict decides. Stripping the learned content out of that loop — keeping the same mechanisms and removing what had been acquired — *changed nothing whatever.* The two arms did not merely score alike; they solved the *identical set of tasks*, and the difference has been measured repeatedly at zero.

So in that domain the capability sat on the enumerator and the verdict, not on what was held. That is a real limit on the reach of this account and the honest reading is narrow: adaptation is not the only route to competence, and where search is cheap and verification is exact, search will do the work and holding will not pay. What the case does not show is that adaptation failed — the ladder was available and the domain did not need it, which is a different finding and a less flattering one to report than a failure would be, because it suggests the mechanism can be present and idle without anything indicating so.

The distinguishing prediction follows from that. *Adaptation should pay where verification is expensive or unavailable*, since that is where search cannot substitute for it, and should not pay where a candidate can be cheaply checked. A domain in which holding helps and verification is cheap would be evidence against this reading; one in which holding never helps anywhere would be evidence against the account.

7 Prior positions

None of the machinery here is new and it should not be dressed as though it were. Four literatures own it, and what this account adds is an arrangement rather than an operation.

Anti-unification. The operation is Plotkin (1970)'s least general generalisation: given two structures, produce the most specific structure of which both are instances. That is exactly rung two and rung three read backwards. *What differs is what is kept.* The classical formulation returns the generalisation and discards the substitutions that recover the inputs from it; here the substitutions are the point, because *the witness is the transform* and applying it elsewhere is the whole operation. An lgg that throws away its witness cannot adapt anything.

Structure-mapping. Gentner (1983) established that analogy runs on relational structure rather than surface features, and that systematic relational systems are preferred to isolated matches — both of which this account assumes rather than argues. *What differs is the purpose.* Structure-mapping is a theory of comparison: what makes two things analogous and how a person aligns them. This is a theory of acquisition, in which the comparison is instrumental — the alignment exists to yield a transform that mints a competence the system did not have. Gentner's theory answers what an analogy *is*; the question here is what one is *for*.

Analogical proportion. The four-term form, a is to b as c is to d, has a developed logical treatment (Prade and Richard, 2014), and parallelogram completion is that form with d unknown. Taken directly and without modification.

Embedding arithmetic, and the reason its critique does not transfer. The word *parallelogram* comes from vector offsets in learned embedding spaces (Mikolov et al., 2013), and that method's evaluation has been substantially criticised: Levy and Goldberg (2014) showed its behaviour is largely accounted for by similarity terms rather than by the offset, and Linzen (2016) showed the standard benchmark is confounded by nearest-neighbour baselines that need no analogy at all.

Those criticisms are correct and they do not apply here, for a reason that is a real difference and not a defence. The offset method is metric: it returns the nearest point to a computed location, so it always returns something, and the question is only whether the something is right. The operation here is symbolic and exact — anti-unification either yields a witness that is well-defined and lossless, or it fails and yields nothing. *There is no nearest neighbour to fall back on and no threshold to tune*, which removes the failure mode the critique identifies and replaces it with a different one: refusing cases a metric method would have answered. That is the cost of the exchange and it is the honest form of it.

8 Falsifiability commitments

1. [SHAPE] **Shape matching and analogy are one operation.** Fails if a case of one cannot be expressed as a case of the other — specifically, if some analogical completion has no formulation as a transform between held signatures, or some shape adaptation has no formulation as an anti-unification with its witness retained.

2. [SHAPE] **The ladder's order is forced, not chosen.** Fails if reordering the rungs leaves the outcomes unchanged on cases where the rungs disagree, which would make the order a preference.

3. [DETAIL] **One mint, then recognition.** Fails on a novel case that yields repeated minting under an unchanged codebook, or none at all.

4. [DETAIL] **Coupling forces a dimension.** Fails if a residual coupled across observed dimensions is absorbed by a member rather than a dimension, without loss.

5. [SHAPE] **Adaptation pays where verification is expensive.** Fails if a domain is exhibited in which candidates are cheaply and exactly checkable and held structure nevertheless contributes measurably — or if no domain can be found in which holding contributes at all.

6. [SHAPE] **Adaptability is per case, not a rank.** Fails if a single ordering of systems by adaptability can be constructed that survives changing the held set.

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